[Week 13] Group Assignment

Team 9: 董苡恩, 范容瑤, 高詩怡

Task 1: Build a Simple Agent and Apply It to Our Data

Task 2: Five Ways Agents and Tools Could Benefit Our Final Project

Our final project applies text mining to TSMC's 2024 Sustainability Report. The current pipeline moves from role based TF-IDF classification to SDG based classification mapped onto the nine SDGs that TSMC reports, and presents the results through an interactive Streamlit dashboard. The five examples below describe how an agent and tool architecture, following the design we studied this week, could extend that pipeline. We do not implement them here; we explain the benefit and the design logic for each.

A common thread runs through all five examples. Each of our existing analysis steps, such as TF-IDF scoring, SDG mapping, and dashboard querying, can be wrapped as a tool, and a selector agent can decide when to call it while an executor agent runs it and returns structured output. This mirrors the slide deck principle of writing a function first and then wrapping the function as a tool.

1. SDG Classification Agent

Tool type: external service plus function wrapper.

Our SDG classification currently relies on TF-IDF combined with a fixed keyword map. We could replace this with an agent paired with a search_sdg_target tool. Given a paragraph from the report, the tool retrieves the official target definitions for the 17 SDGs, and the agent then judges by meaning which SDG the paragraph fits best and returns a confidence score. The benefit is that classification no longer depends on a static keyword list, so passages that express a goal in different words can still be classified correctly. This reflects the lecture point that a tool converts unstructured text into structured JSON output.

2. Cross-Company Benchmark Agent

Tool type: external service, with a selector and executor pair.

This extends the cross-company and longitudinal ESG analysis we proposed. We could design an agent with a fetch_report tool that takes a company name and a year, retrieves the matching sustainability report from a corpus, runs the same TF-IDF and SDG pipeline, and returns structured results. A selector agent decides which companies to compare, for example TSMC against UMC against Samsung, while an executor agent performs the retrieval and analysis and assembles a comparison table. The benefit is that a workflow that currently handles one report at a time becomes a repeatable framework for comparison across firms and across years.

3. Claim Verification Agent

Tool type: external resource, following the news lookup example in the slides.

This example closely follows the teacher agent that looks up news from the Taipei Times. The agent extracts a sustainability claim from the report, such as a net zero target by 2050 or a stated share of renewable energy use, then uses a search_news tool to find related coverage from external news sources and compares the claim against that coverage to flag possible greenwashing. The benefit is that the analysis moves beyond internal word counts and becomes a cross check between what the report states and what external evidence shows, which is especially valuable for ESG research.

4. Expert and Framework Citation Agent

Tool type: external service, modeled on AG_search_expert (knows who) and AG_search_textbook (knows what).

When the analysis touches a specific topic, for example Scope 3 emissions or supply chain management, the agent calls a search_framework tool to find the authoritative frameworks that apply, such as the GHG Protocol, the CSRD, and Taiwan Corporate Governance 3.0, together with the relevant scholars. It then attaches these references to the analysis output. The benefit is that our text mining results are no longer only word frequencies; they connect to established academic and regulatory frameworks, which strengthens the credibility of our academic writing.

5. Interactive Q&A Agent for the Dashboard

Tool type: a ConversableAgent connected to our analysis database.

We already built a Streamlit dashboard and a QR code page for exhibition use. We could add a ConversableAgent that lets visitors ask questions in natural language, for example how much attention TSMC gives to water resource management, where the agent uses a query_tfidf_result tool to look up our existing TF-IDF and SDG analysis and answer. This demonstrates the lecture idea that a tool is an adapter connecting an agent to other digital artifacts, which in this case is our own results database. The benefit is that a static set of charts becomes a conversational exhibit, which greatly increases interaction for visitors at the showcase.

Summary

The table below maps each example to its tool type and its main benefit for our project.

Example	Tool type	Main benefit
SDG Classification Agent	External service plus function wrapper	Semantic classification replaces a fixed keyword map
Cross-Company Benchmark Agent	External service, selector and executor pair	Turns single report analysis into a repeatable comparison framework
Claim Verification Agent	External resource (news lookup)	Cross checks report claims against outside evidence to flag greenwashing
Expert and Framework Citation Agent	External service (who and what lookup)	Links results to academic and regulatory frameworks
Interactive Q&A Agent	ConversableAgent on results database	Makes the dashboard a conversational exhibit